

Population Distributions of Footbridge Modal Properties: A PRISMA-Guided Review and Hierarchical Bayesian Analysis

Samadhi Loku Hetti Arachchilage^a, Colin Caprani^a, Stana Živanović^b

^a*Department of Civil Engineering, Monash University, Melbourne, Victoria, Australia*

^b*School of Engineering, University of Warwick, Coventry, United Kingdom*

Abstract

Keywords:

1. Introduction

The increasing adoption of light weight materials and slender structural forms has made modern footbridges more susceptible to human-induced vibrations. As a result, vibration serviceability under pedestrian loading has become an important design consideration to ensure acceptable structural performance and user comfort. Such assessment depends on several aspects of the bridge, including its structural form, material properties, and dynamic characteristics [1, 2, 3]. Therefore, prior knowledge or reliable estimation of key modal parameters is essential at the design stage, as these parameters provide an initial basis for assessing the feasibility of a proposed design [4, 5, 6, 7].

The current vibration serviceability codes and guidelines represent pedestrian loading through simplified load models to enable practical application at the design stage [8, 9, 10, 11, 12, 13, 14]. Since these load models are used to assess the vibration response of a wide range of footbridges, their adequacy should be evaluated in terms of the resulting structural response, rather than solely through how accurately they reproduce pedestrian loading.

The key limitation of many existing simplified load models is that pedestrians are represented as moving forces, with the corresponding load models calibrated without explicitly accounting for the dynamic interaction between the pedestrians and the supporting structure [15, 16, 17]. Human–structure interaction (HSI) alters the dynamic properties of the coupled pedestrian–structure system, and can lead to structural responses that differ substantially from those predicted by simplified load models. These effects can be

31 particularly important for lightweight structures and for cases with relatively
32 large pedestrian-to-structure mass ratios [18, 17]. Consequently, simplified
33 code models calibrated without accounting for HSI may exhibit systematic
34 response bias that varies with the dynamic characteristics of the bridge, mo-
35 tivating their reassessment and response-based recalibration.

36 Response-based calibration provides such a framework by comparing the
37 response predicted by the simplified design model with that obtained from ex-
38 perimental observations or a higher-fidelity reference model over a prescribed
39 range of structural properties. Previous studies have adopted this approach
40 by evaluating simplified pedestrian loading or vibration serviceability mod-
41 els over ranges of natural frequency, damping and other structural param-
42 eters [19, 3, 17, 18, 20]. However, these structural domains considered in
43 such studies have generally been defined using prescribed parameter ranges,
44 idealised structural configurations, or regularly sampled parameter combi-
45 nations rather than using the observed statistical distribution of dynamic
46 properties within the footbridge population. As a result, some structural
47 conditions may receive disproportionate importance in the calibration sim-
48 ply because of the adopted sampling scheme, rather than because they are
49 common in practice.

50 For code-model calibration, it is important to know not only the range
51 of structural properties that may occur, but also how common these proper-
52 ties are within the footbridge population. Several recent studies have made
53 valuable efforts to compile databases of footbridge characteristics and modal
54 properties. Oliveira et al. (2014) [21] reported fundamental frequencies
55 for 79 footbridges in Portugal, while Van Nimmen (2015) reviewed data
56 for more than 160 footbridges [22, 23]. Wei et al. (2019) [24] assembled
57 dynamic-property data for 138 footbridges, and Gong et al. (2024) [25]
58 more recently studied a collection of footbridges in China. These studies
59 provide a substantial starting point, but their datasets were assembled for
60 different purposes, report different properties, and include some of the same
61 bridges. Further observations are dispersed across many individual case stud-
62 ies [26, 6, 27, 20]. Consolidating these records, identifying duplicate bridges,
63 and recording which modal properties are available would provide a clearer
64 basis for estimating their observed distributions.

65 This study addresses the above gap in two stages. First, a systematic
66 literature search and screening procedure following the PRISMA framework
67 is used to identify relevant publications and compile bridge-level data on key
68 modal properties. Second, hierarchical Bayesian regression is used to proba-
69 bilistically characterise incompletely observed modal properties while retain-
70 ing the uncertainty associated with their estimation. The resulting database
71 provides a population-level description of key footbridge modal properties

and their relationships, allowing the prevalence of different combinations of natural frequency, damping ratio and modal mass to be quantified. Such information provides a statistical basis for distinguishing commonly occurring structural conditions from those that are comparatively rare. This, in turn, can support the selection and weighting of representative structural parameter combinations used in vibration-serviceability assessment and code-model calibration.

Accordingly, this study aims to characterise the modal properties of the footbridge population using a systematically assembled bridge-level database of approximately 250 structures. The principal contributions of the study are: (i) the systematic identification and consolidation of published footbridge dynamic-property data; (ii) the construction of a large, non-duplicated database of bridge-level modal properties; (iii) the probabilistic treatment of incompletely observed damping and modal-mass information using hierarchical Bayesian regression; and (iv) the population-level characterisation of modal properties relevant to vibration-serviceability assessment and code-model calibration.

2. Systematic database development

3. Bayesian treatment of incomplete bridge data

4. Discussion

5. Conclusions

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